

The Double-Slit Delayed-Choice Experiment: A Classical Innovation

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The foundations and thus the strangest notions in quantum physics lie within the model's divergences from the classical theory of light. Where certain experiments may appear to be inherently quantum mechanical in nature, beyond surface level they can be explained using purely classical theory. We explore the "Double-Slit Experiment" with an adaptation from Wheeler's "Delayed-Choice Experiment". We remodel and propose an analogue experiment that retrieves the time-measured double-slit results including interference patterns and which-path information using Jones calculus and its classical analysis.

I. Background

– Motivation –

Quantum mechanics rises in areas where a classical theory of light cannot comprehensively describe the relevant phenomena. In the "classical" theory, light can be described as the propagation of linearly independent electric and magnetic wave components in space and time. Where the classical interpretation diverges, the quantum theory provides a satisfactory explanation for various phenomena that otherwise could not be explained. The dependency of measurement outcomes on the observer's role in quantum mechanics shows fundamental discrepancies between the two theories that must be investigated. As such, many quantum interpretations appear to be different to that of the classical theory. We attempt to answer the following questions: *Can one decompose non-classical predictions to their core to unveil a simpler explanation compared to what quantum mechanics offers? Which physical phenomena can be investigated? Will the predictions agree in their relative causal structures?*

In this paper, we consider the "Delayed-Choice Double-Slit experiment" by proposing an analogous experiment to adapt the formerly investigated quantum model By Pantaleoni (2016) [5] with purely classical theory. We compare our findings, highlighting areas where similar results are retrieved, areas where differences lie, and the structural differences between the two approaches.

– The Double-Slit Experiment –

First investigated by Thomas Young, the "Double-Slit Experiment" (DSE) is a notable demonstration of the wave-particle duality of light. Although many variations of the experiment exist, the double-slit experiment involves a coherent, monochromatic light source incident

on a barrier with two sufficiently narrow slits, as well as a screen positioned behind the barrier which records the incident pattern of light on it (FIG.1). In a *Fraunhofer regime* $L \gg \delta$, where δ is the slit width and L is the distance between the barrier and the screen. [2] For a Gaussian laser light source with a short pulse, the experiment can be investigated using the two models. Light can classically be modelled as an electromagnetic wave propagating towards the barrier. For a fixed direction r , this is denoted as $E(t) = E_0 e^{i(k_0 r - \omega t + \phi)}$ where $k_0 = \frac{2\pi}{\lambda}$ and ϕ are respectively the free-space wavevector, and phase term. Given a wave function $E(t)$, the intensity at a point is a function of a phase difference ϵ between the waves from each slit. Hence the intensity is

$$I = \|E(t) + E(t)e^{i\epsilon}\|^2 \quad (1)$$

$$= (E(t) + E(t)e^{i\epsilon})(E^*(t) - E^*(t)e^{i\epsilon}) \quad (2)$$

$$= E_0^2(2 + 2 \cos \epsilon) \quad (3)$$

After diffracting through the slits, the two new propagating wavefronts interfere constructively/destructively to form bright/dark fringes on the screen. The path difference between the waves $\Delta s = m\lambda$ where λ denotes the light wavelength is equivalent to the phase difference of light due to a delayed time of arrival to a point on the screen which causes the resultant double-slit pattern. For $m = 1, 2, 3, \dots$ this yields:

$$\Delta s = \begin{cases} m\lambda & \text{constructive} \\ (m - \frac{1}{2})\lambda & \text{deconstructive} \end{cases}$$

Given a single photon source, the wave-like behaviour of photons is unexpected. Considering a quantum analysis, before placing a detector in front of a slit, the photon is in a superposition at the two slit locations x_1 and x_2 .

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|x_1\rangle + |x_2\rangle) \quad (4)$$

Delocalisation in the momentum distribution of light projected onto the detection screen results in an interference pattern.

$$\langle \hat{p} | \psi \rangle = \frac{1}{\sqrt{2}} (\langle \hat{p} | x_1 \rangle + \langle \hat{p} | x_2 \rangle) \quad (5)$$

$$= \frac{1}{\sqrt{2}} (F_\delta(x_1) + F_\delta(x_2)) \quad (6)$$

where $\hat{p}$ is the momentum operator, δ is the finite width of each slit, and $F(x_i)$ is the Fourier transform at slit i onto the momentum space [4]. If a measurement is made on a position basis, information about this distribution will disappear as position and momentum are not commuting and the particle pattern is retrieved. The **which-path information (WPI)** is the information available regarding the path that light takes through either slit. WPI is correlated to the availability of the interference pattern. In FIG.1(a), the travelled paths are indistinguishable and interference information is available. In FIG.1(b) the contrary is true and WPI is available and distinguishable [6]. The combination of classical and quantum models provides a sufficient but disjoint interpretation of the experiment.

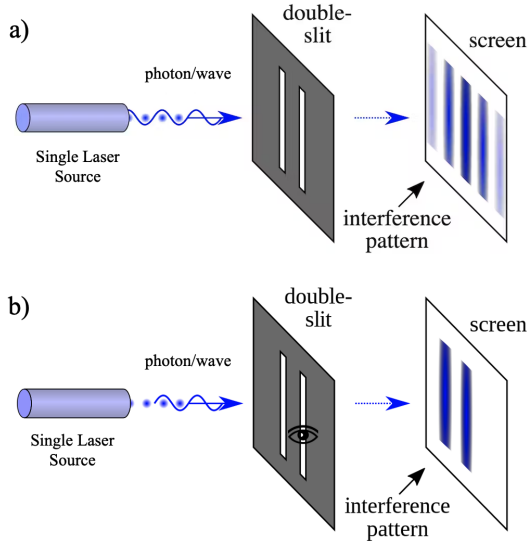


FIG. 1: Simplified schema of the DSE with **a)** no detector in place (WPI available) and **b)** detector in front of a slit (WPI unavailable) (Image source: Johannes Kalliauer (2017))

– Wheeler’s Delayed-Choice –

In the 1980s John Archibald Wheeler proposed a *Gedanken* experiment to investigate the causal significance of the observer in the measurement collapse [7]. In a variation of the proposal, coherent and monochromatic light is incident on a Mach-Zehnder interferometer setup. The incident mode is split into reflected and transmitted components via the first beam-splitter

which travel length-identical paths. The observer is then granted a “choice” to place a second beam-splitter or not before the two light detectors (FIG.2). The experiment is analogous to the DSE in nature and demonstrates the wave-particle duality of light depending on the presence of the second beam-splitter.

In the case where the second beam-splitter is not present, detection at each splitter is detected 50% of the time with no simultaneous detection. Wheeler interprets this as the light “photon” either probabilistically reflecting or transmitting at the first beam-splitter, then travelling through one of the arms and thus yielding the detection result. Hence the placement choice results in a *particle* behaviour. In the case where the beam-splitter is present, detection at only one detector is observed. This is interpreted as EM-waves splitting to transmission and reflection modes upon arriving at the first beam splitter. Reflection modes change in phase and thus two arms host light of differing phases. After arriving at the second beam-splitter, deconstructive interference of light results in no detection at one detector, while constructive interference results in detection. Denote that the type of detection here is not strictly defined and is subject to the experiment. Wheeler interprets this as a *wave* behaviour of light.

The “delayed choice” for the presence of the removable beam-splitter poses a potential causal problem. Given the lack of the beam-splitter, one may assume that light will behave as a particle. However, if the observer inserts the beam-splitter *after* the light has entered the system, a wave detection pattern is observed. One can evaluate this result to be retrocausal as the entered photon would require to move back in time to change its subsequent behaviour [1]! But can the observations be explained without any reliance on retrocausal structures?

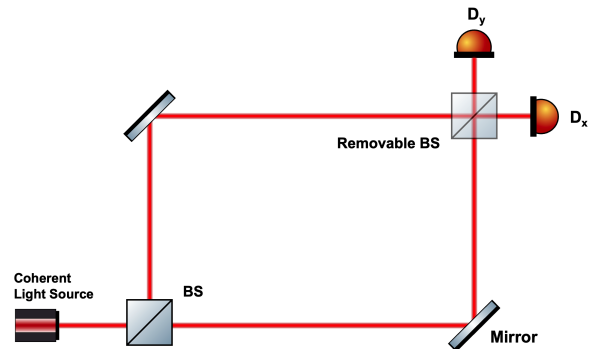


FIG. 2: A variation of Wheeler’s thought experiment with the top-right beam-splitter determining the delayed-choice behaviour of light.

II. Approach

– The Proposed Jones Calculus –

For the light propagating through the system, we consider a Jones Vector representation for the polarisation state of light. The orthogonal polarisation directions are defined with the basis $\mathcal{P} = \{|H\rangle, |V\rangle\}$ [1] where $|H\rangle$ and $|V\rangle$ represent orthogonal horizontal and vertical directions of polarisation respectively. The Cartesian direction of light propagation in horizontal/vertical arms (orthogonal x, y directions) regardless of the sign is defined by the basis $\mathcal{D} = \{|X\rangle, |Y\rangle\}$. The tensor product on the space of polarisation and propagation yields a basis for the elements of $\mathcal{B}$ in $\mathbb{C}^4$ as such:

$$\mathcal{P} \otimes \mathcal{D} = \mathcal{B} = \{|H\rangle|X\rangle, |H\rangle|Y\rangle, |V\rangle|X\rangle, |V\rangle|Y\rangle\} \quad (7)$$

where

$$\begin{aligned} |H\rangle|X\rangle &= [1 \ 0 \ 0 \ 0]^T \\ |H\rangle|Y\rangle &= [0 \ 1 \ 0 \ 0]^T \\ |V\rangle|X\rangle &= [0 \ 0 \ 1 \ 0]^T \\ |V\rangle|Y\rangle &= [0 \ 0 \ 0 \ 1]^T \end{aligned}$$

The purpose behind this basis definition is to define each optical element's operation (Appendix VII A) on light using matrices $Mat_{4 \times 4}(\mathbb{C})$ acting on a Jones Vector representation of input light (FIG.3).

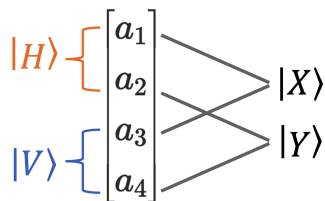


FIG. 3: Coordinates representation in the defined basis $\mathcal{B}$ of polarisation and propagation direction. For $i \in [1, 4]$, $a_i \in \mathbb{C}$.

– The Proposed Analogue –

To construct a delayed-choice analogue experiment, we are motivated by a setup from Pantaleoni G. (2017). As shown in FIG.4, a Gaussian beam is emitted from the laser source, propagating through the system at two different paths incident on a diagonal polariser at 45° . The light is then incident on a beam-splitter (BS) which splits light into distinctive *short* and *long* paths (straight down, and reflected from mirrors). After arriving at a polarised

beam-splitter (PBS1), some light components reach a *Quarter Waveplate* (oriented at θ°), *Send-Back Mirror*, *Quarter Waveplate* setup (QWP-SBM-QWP) and are then reflected back to PBS1. Components then travel once more to BS, where some components arrive at a *Variable Waveplate* (VWP) imparting a variable phase ε only to the $|V\rangle$ component. The light is then incident on a *Half Waveplate* (HWP) oriented at 22.5° which constructs diagonal/anti-diagonal light. Upon a phase shift, light is finally incident on another polarised beam-splitter PBS2 which leads to detection at two orthogonal detectors.

– Pre-QWP –

The light becomes diagonally-polarised ($(|H\rangle + |V\rangle)|Y\rangle \rightarrow |+\rangle|Y\rangle$) after the polariser. The proposed light source emits a Gaussian envelope with the function $f(t, \sigma, l) = \exp(-\frac{(ct-l)^2}{2\sigma^2})$ where c and σ are the speed and standard deviation of light respectively and l is the distance from the polariser, resulting in the interferometer's input light as

$$E(t) = f(t, \sigma) |+\rangle|Y\rangle. \quad (8)$$

The beams-splitter BS equally splits incident intensities, applying $\frac{\pi}{2}$ phase shift in the reflected mode (Appendix 1). The overall (time-independent) polarisation state is then

$$\vec{\psi} = \frac{1}{\sqrt{2}} |+\rangle(|X\rangle + i|Y\rangle) \quad (9)$$

Given there is path length difference between the short and long arm (FIG.4), the component $\frac{1}{\sqrt{2}} |+\rangle|X\rangle$ will arrive at PBS1 with a time delay $\tau = 2d/c$ where d is the distance from BS to the first mirror. By definition, the polarising beam-splitter PBS1 only reflects $|V\rangle$ (with another phase gain) and transmits $|H\rangle$. This results in a loss of half of the intensity that arrives from each arm, as short (long) propagation leads to a loss of vertical (horizontal) component which exits the system. Hence after PBS1

$$\frac{1}{\sqrt{2}} |+\rangle|Y\rangle \rightarrow \frac{1}{2} |H\rangle|Y\rangle \quad (10)$$

$$\frac{i}{\sqrt{2}} |+\rangle|X\rangle \rightarrow -\frac{1}{2} |V\rangle|Y\rangle \quad (\text{after time } \tau) \quad (11)$$

$$\therefore \vec{\psi} = \frac{1}{2} (|H\rangle - |V\rangle)|Y\rangle \quad (12)$$

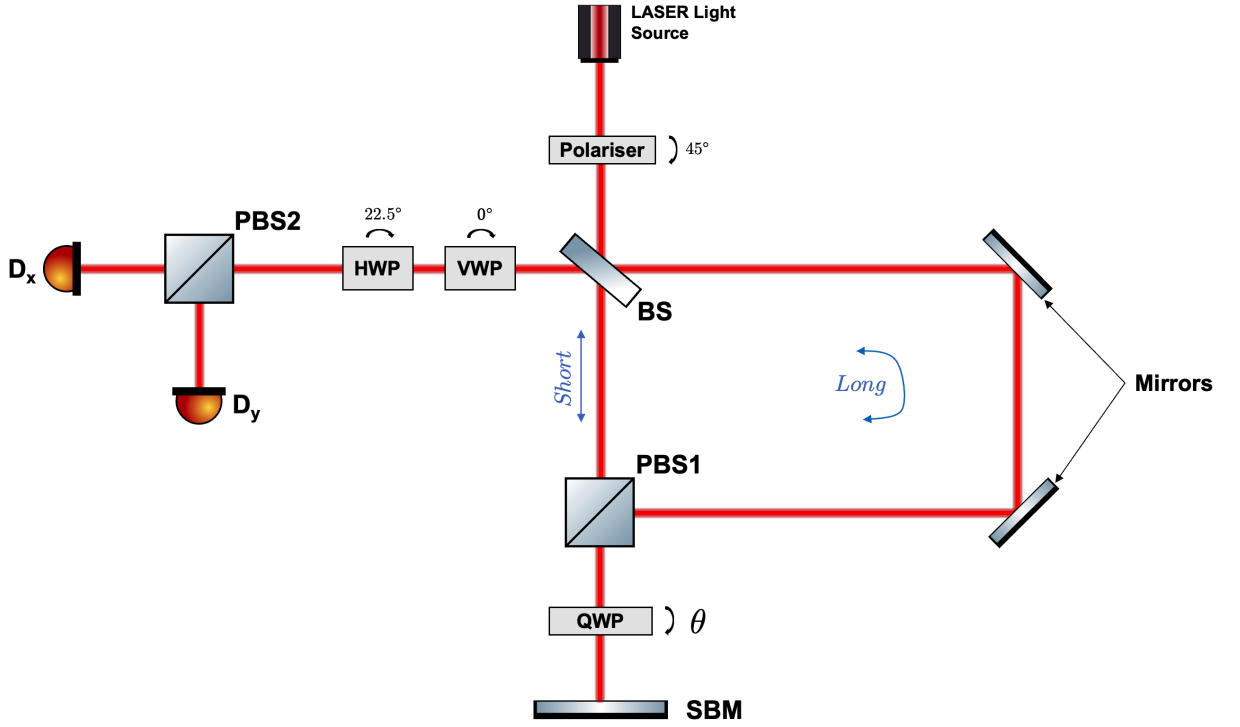


FIG. 4: The overall schema of the analogue setup. The red lines represent the propagation of light through the interferometer arms. Angles shown for Polariser, QWP and HWP are with respect to the horizontal polarisation axis. The detectors on the left are orthogonal and identical in path length to PBS2. The *Short* and *Long* paths are demonstrated.

– QWP-SBM-QWP –

The *Quarter Waveplate, Send-Back Mirror, Quarter Waveplate* setup is a field standard for polarisation state switching. The presence (absence) of the QWP is analogous to Wheeler’s presence (absence) of the removable beam-splitter. When oriented at an angle θ about the horizontal polarisation axis, the light is imposed to a $\frac{\pi}{2}$ phase shift, the SBM acts to reflect the light by another phase gain of π on the vertical component, and the light is again incident on the QWP at an angle $\pi - \theta$ (due to reflection). This acts to swap the singular $|H\rangle$ and $|V\rangle$ ’s arm paths with *Short* ($|H\rangle$) $\leftrightarrow$ *Long* ($|V\rangle$). This yields

$$\text{Short } |H\rangle \longrightarrow -\sin(2\theta) |V\rangle \quad (13)$$

$$\text{Long } |V\rangle \longrightarrow \sin(2\theta) |H\rangle \quad (14)$$

after re-entering PBS1 and respectively reflecting/transmitting. Ignoring the intensity loss, this implies that the polarisation state at each traversed arm is now reversed, and hence the components will arrive back at BS at the same time as the time delay τ is reversed. The light taking the *Short-Long* and *Long-Short* no longer contains distinguishable WPI due to the superposition of the two paths which is analogous to the presence of the double-slit pattern in the DSE.

Substantially, the rotation of QWP along its optical axis is equivalent to adjusting the intensity of light at a point on the DSE screen with max/min intensities corresponding with the presence of WPI.

It is known that for specific θ values, the path distinguishability is maximised. Notably, this occurs when the $|H\rangle$ and $|V\rangle$ components respectively take *Short-Short* and *Long-Long* paths as the *Long-Long* path will arrive with a time delay of 2τ ^[2]. In this configuration the presence of time-dependent WPI is analogous to applying a measurement on one of the slits in the DSE, and hence losing the momentum space projection on the screen. The polarisation states of light until after the second passage through PBS1 ($|V\rangle$ component phase gain) in time-order are:

$$\vec{\psi}_{\text{Short-Short}} = \frac{1}{2} |H\rangle |Y\rangle \longrightarrow \frac{\cos 2\theta}{2} |H\rangle |Y\rangle \quad (15)$$

$$\vec{\psi}_{\text{Short-Long}} = \frac{1}{2} |H\rangle |Y\rangle \longrightarrow -\frac{i \sin 2\theta}{2} |V\rangle |X\rangle \quad (16)$$

$$\vec{\psi}_{\text{Long-Short}} = -\frac{1}{2} |V\rangle |Y\rangle \longrightarrow -\frac{\sin 2\theta}{2} |H\rangle |Y\rangle \quad (17)$$

$$\vec{\psi}_{\text{Long-Long}} = -\frac{1}{2} |V\rangle |Y\rangle \longrightarrow -\frac{i \cos 2\theta}{2} |V\rangle |X\rangle \quad (18)$$

One can importantly notice that for $\theta = (2k-1)\frac{\pi}{4}$, $k \in \mathbb{Z}$ the *Short-Short* and *Long-Long* polarisation states vanish, while for $\theta = k\frac{\pi}{2}$ the *Short-Long* and *Long-Short* states vanish. Thus the rotation of QWP ultimately determines the state of which-path information and hence the interference pattern.

– Interference Detection –

Upon ‘polarisation swapping’, PBS1 splits each orthogonal component by transmitting $\vec{\psi}_{Long-Short}$ and reflecting and reflecting $\vec{\psi}_{Short-Long}$ with another $\frac{\pi}{2}$ phase gain as shown in eq. (13) & (15).

With the time-delay between $\vec{\psi}_{Long-Short}$ and $\vec{\psi}_{Short-Long}$ now resolved, the states are incident at BS simultaneously while $\vec{\psi}_{Short-Short}$ and $\vec{\psi}_{Long-Long}$ states are distinguishable in time as discussed before. At BS, intensity loss in two components of light (i.e. $|V\rangle$ from $\vec{\psi}_{Short-Long}$ and $|H\rangle$ from $\vec{\psi}_{Long-Short}$) equates to $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$. This leads to an overall (and final) intensity efficiency of $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ for the system. The polarisation states in time-order after BS are:

$$\vec{\psi}_{Short-Short} = \frac{i \cos 2\theta}{2\sqrt{2}} |H\rangle |X\rangle \quad (19)$$

$$\vec{\psi}_{Short-Long} = -\frac{i \sin 2\theta}{2\sqrt{2}} |V\rangle |X\rangle \quad (20)$$

$$\vec{\psi}_{Long-Short} = -\frac{i \sin 2\theta}{2\sqrt{2}} |H\rangle |X\rangle \quad (21)$$

$$\vec{\psi}_{Long-Long} = -\frac{i \cos 2\theta}{2\sqrt{2}} |V\rangle |X\rangle \quad (22)$$

where we notice that $\vec{\psi}_{Short-Long} = \vec{\psi}_{Long-Short}$ as expected.

Arriving at the VWP set parallel to the horizontal polarisation axis, the polarisation states undergo a variable phase gain ε to their $|V\rangle$ such that $|V\rangle |X\rangle \rightarrow e^{i\varepsilon} |V\rangle |X\rangle$ (See Appendix 1 VII A). The

polarisation states in time-order are now:

$$\vec{\psi}_{Short-Short} = \frac{i \cos 2\theta}{2\sqrt{2}} |H\rangle |X\rangle \quad (23)$$

$$\vec{\psi}_{Short-Long} = -\frac{i \sin 2\theta}{2\sqrt{2}} e^{i\varepsilon} |V\rangle |X\rangle \quad (24)$$

$$\vec{\psi}_{Long-Short} = -\frac{i \sin 2\theta}{2\sqrt{2}} |H\rangle |X\rangle \quad (25)$$

$$\vec{\psi}_{Long-Long} = -\frac{i \cos 2\theta}{2\sqrt{2}} e^{i\varepsilon} |V\rangle |X\rangle \quad (26)$$

The HWP oriented at $\frac{\pi}{8}$ causes incident $|H\rangle |X\rangle \rightarrow |+\rangle |X\rangle$, and $|V\rangle |X\rangle \rightarrow |-\rangle |X\rangle$ where $|-\rangle$ is anti-diagonal polarisation. Finally, PBS2 operates similar to PBS1 to separate orthogonal polarisation components where all $|H\rangle$ and $|V\rangle$ components are detected by D_x and D_y respectively. Of course, the reflection phase gain must again be considered.

The final polarisation states can be evaluated on a detector basis in time-order. At D_x :

$$\vec{\psi}_{Short-Short} = \frac{i \cos 2\theta}{4} |H\rangle |X\rangle \quad (27)$$

$$\vec{\psi}_{Short-Long} = -\frac{ie^{i\varepsilon} \sin 2\theta}{4} |H\rangle |X\rangle \quad (28)$$

$$\vec{\psi}_{Long-Short} = -\frac{i \sin 2\theta}{4} |H\rangle |X\rangle \quad (29)$$

$$\vec{\psi}_{Long-Long} = -\frac{ie^{i\varepsilon} \cos 2\theta}{4} |H\rangle |X\rangle \quad (30)$$

And at D_y :

$$\vec{\psi}_{Short-Short} = \frac{\cos 2\theta}{4} |V\rangle |Y\rangle \quad (31)$$

$$\vec{\psi}_{Short-Long} = -\frac{e^{i\varepsilon} \sin 2\theta}{4} |V\rangle |Y\rangle \quad (32)$$

$$\vec{\psi}_{Long-Short} = \frac{\sin 2\theta}{4} |V\rangle |Y\rangle \quad (33)$$

$$\vec{\psi}_{Long-Long} = -\frac{e^{i\varepsilon} \cos 2\theta}{4} |V\rangle |Y\rangle \quad (34)$$

III. Discussion

– Analysis –

The correspondence between the proposed analogue and the classical model lies within achieving similar interference and WPI results while maintaining a cohesive causal structure. Recall that the *Short-Long* and *Long-Short* configurations have a similar total distance and so the polarisation states through each configuration will arrive in phase at the detectors. The total wave intensities

[1] Notice that the use of Dirac notation here is purely for contextual reasons and the basis throughout the analysis remains faithful to the classical model.
[2] Given experimental values for ω and hence τ , the overlap for the intensity Gaussians will be minimised.

at time t given by $I_x(t)$ and $I_y(t)$ at each detector can be described as the multiplication of the Gaussian function and the evolved polarisation states for all configurations.

$$\begin{aligned}
I_x &= \|f(t, \sigma, l)\vec{\psi}_{Short-Short} \\
&+ f(t, \sigma, l + 2d)(\vec{\psi}_{Short-Long} + \vec{\psi}_{Long-Short}) \\
&+ f(t, \sigma, l + 4d)\vec{\psi}_{Long-Long}\|^2 \\
&= \|f(t, \sigma, l)\frac{i \cos 2\theta}{4} \\
&- f(t, \sigma, l + 2d)\frac{i \sin 2\theta}{4}(1 + e^{i\varepsilon}) \\
&- f(t, \sigma, l + 4d)\frac{\cos 2\theta}{4}e^{i\varepsilon}\|^2
\end{aligned}$$

$$\begin{aligned}
I_y &= \|f(t, \sigma, l)\vec{\psi}_{Short-Short} \\
&+ f(t, \sigma, l + 2d)(\vec{\psi}_{Short-Long} + \vec{\psi}_{Long-Short}) \\
&+ f(t, \sigma, l + 4d)\vec{\psi}_{Long-Long}\|^2 \\
&= \|-f(t, \sigma, l)\frac{\cos 2\theta}{4} \\
&+ f(t, \sigma, l + 2d)\frac{\sin 2\theta}{4}(1 - e^{i\varepsilon}) \\
&- f(t, \sigma, l + 4d)\frac{\cos 2\theta}{4}e^{i\varepsilon}\|^2
\end{aligned}$$

where the variables are defined as before and the polarisation states are used from the final detector outputs. Importantly, the intensity equations determine that θ impacts the intensity of each polarisation state $\vec{\psi}$ and ε from the VWP impacts the phase difference for each path.

– DSE Comparison –

Upon understanding the role of ε and θ we may derive similarities between the DSE and the analogue. The role of ε , as discussed, is to input a relative phase between the polarisation and hence between the different paths. $\Delta\varepsilon$ can then be used to change the interference pattern through its peak intensity (Fig.5). Recall that for the no WPI case the total detection (i.e. $\vec{\psi}_{Short-Long} + \vec{\psi}_{Long-Short}$) of the polarisation states at the detectors are:

$$\begin{aligned}
D_x &: -\frac{i \sin 2\theta}{4}(1 + e^{i\varepsilon})|H\rangle|X\rangle \\
\Rightarrow I_{peak,x} &= \frac{\sin^2 2\theta}{8}(1 + \cos \varepsilon) \\
D_x &: \frac{\sin 2\theta}{4}(1 - e^{i\varepsilon})|V\rangle|Y\rangle \\
\Rightarrow I_{peak,y} &= \frac{\sin^2 2\theta}{8}(1 - \cos \varepsilon)
\end{aligned}$$

where the intensities I are derived from the complex conjugates. Notice that the intensity equations are very similar to that of eq. (1-3). Also, the peak intensities at both detectors are shifted in phase ε by a factor of π . The change in the amplitude term involving θ as well as modulation in the phase involving ε exactly replicates the double slit interference here! Hence it can be stated that with unique sets of (θ, ε) the interference and hence the WPI can be reconstructed.

– Which Path Information –

Comparing the theoretical results to the experimental realisation by Pantaleoni (Fig.6) shows that the total peak intensities $I_{peak} = I_x + I_y$ agree with the derivations. Starting from the bottom-right diagram of Fig.6 where no WPI is available shows maximum intensity at the central peak and until the top-left shows WPI with time-distinguishable paths. This can then be summarised as:

$$\begin{aligned}
\theta = k\frac{\pi}{2} &\Rightarrow \vec{\psi}_{S/L-L/S} \Rightarrow \text{WPI Min.} \quad \& \\
\theta = (2k-1)\frac{\pi}{4} &\Rightarrow \vec{\psi}_{S/L-S/L} \Rightarrow \text{WPI Max.}
\end{aligned}$$

where $k \in \mathbb{Z}$ and S, L stand for *Short, Long*.

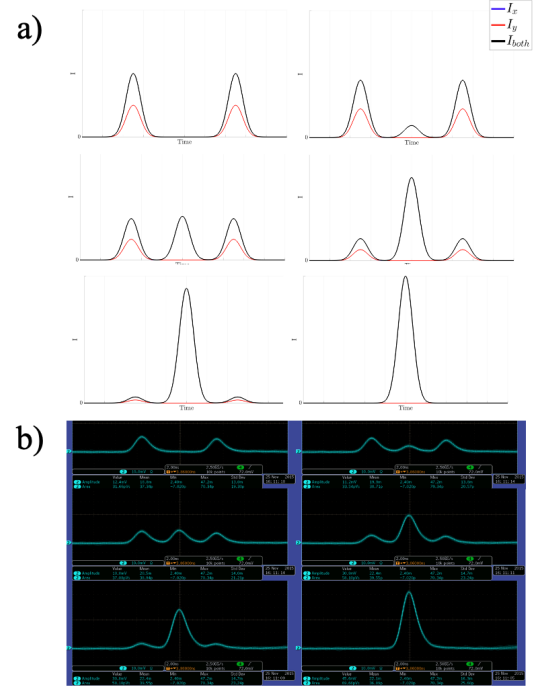


FIG. 6: The output intensities versus time for **a)** the analogue's theoretical results with a range of $\theta = k\frac{\pi}{20}, k \in [0, 5]$. Black line represents the sum intensity, blue line output from D_x , and red line output from D_y . $\varepsilon = 0$ throughout. **b)** The experimental results from Pantaleoni (2017).

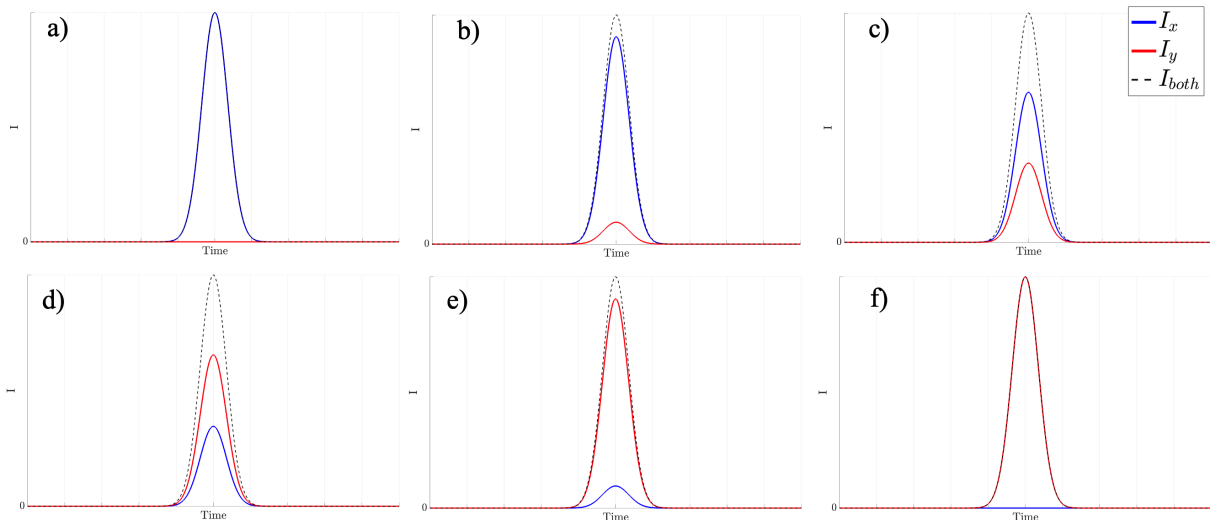


FIG. 5: Intensity against time graphs at the detectors and overall. $\theta = \frac{\pi}{4}$ and alphabetically, $\varepsilon = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi$

– *DCE Ontology* –

We clearly understand that the notion of a delayed-choice is closely tied to the orientation of QWP. Where the WPI is maximised, the double-slit interference pattern is independent of ε and on the contrary where WPI is absent, the $\vec{\psi}$ terms depend on ε . Critically, the classical model predicts that until the delayed choice position, light behaves identically in both arms as different components and can be split into different polarisation states. This is unlike the quantum theory where the passage of a photon through an arm is determined probabilistically as a superposition of all possible eigenstates. Hence the delayed-choice is classically redundant as the light wave has already made the path ‘choice’ from the beginning. This implies a lack of retrocausality due to the indistinguishability of the physical situation before interacting with QWP regardless of the WPI status. Without retrocausality, the delayed-choice notion can be interpreted as follows: “Given that the observation can only occur at the detector in a certain time and space, the ‘choice’ of measurement is only revealed at the detector once the choice has already been made”. The crucial idea that leads to this interpretation is that the notion of the photon’s past is not well defined in classical theory and hence the notion of measurement disturbance is challenged as the information regarding the past is determined **after** the detection. Vice versa, no process can be done in the past to determine the future state of the light.

– *Future Directions* –

Adapting analogous models to unveil the classical modelling under the quantum interpretation of various experiments is an attractive area to study in the future. Regarding the Delayed-Choice Double-Slit Experiment, although a great deal of similarities was observed using the proposed analogue, further classical analysis of the delayed choice counterparts (e.g. quantum erasure) is required. The Quantum eraser experiment [3], requires photon entanglement which is famously incompatible with the classical model. The existence of the Wigner quasiprobability distribution also proposes a classical limit that, in this era, limits the derivation of fully compatible models.

IV. Conclusion

In this paper, we constructed an analogous experiment that adapts the idea of Wheeler’s Delayed-Choice experiment to recreate the Double-Slit experiment’s results without any reliance on quantum theory. Remodelling Pantaleoni (2017) [5] with adjustments using a VWP to impart phase on the interference, we determined that for a unique set of QWP rotation angles θ and VWP phase terms ε , the double-slit patterns can be retrieved. The effective presence of QWP results in a minimisation of WPI and maximisation of interfering Gaussians while the effective lack of QWP results in a maximisation of WPI and minimisation of interfering peaks (time-distinguishable path detections still available). This is shown to be physically analogous to that of the double slit. Regarding the

delayed-choice, we conclude that the classical theory is able to mitigate retrocausality by stating that the light's choice prior to the QWP is not well-defined and after the incidence, it is only manifested at the detectors once the choice and hence the WPI is already determined. Within its causal structure, this analogue offers a simpler explanation for the phenomena at hand, while successfully conforming to experimental results.

V. Acknowledgement

We would like to acknowledge the efforts of Giacomo Pantaleoni for providing their thesis on the subject which was the basis for the experimental confirmations, as well as assisting us through discussions. This acknowledgement should also be extended to Sahand Mahmoodian for providing their lecture notes for PHYS2921 (Optics) and assisting us through various discussions. Finally, we would like to acknowledge Angela Karanjai's kind efforts

as the supervisor of the project for discussing and assisting with various ideas and most importantly for proposing the research idea.

VI. Contribution & Reflection

Throughout the project, the authors both worked together and separately to solve various aspects of the problem. Where executive decisions about the modelling were required, we discussed in person to derive our mathematical tools for the analogue. Separately, we focused on understanding the problem, writing about the problem background, and deriving different proposals for the analogue. The initial part of the project was spent on learning quantum mechanics fundamentals, the middle part was spent on understanding the problem, and the latter part on formulating an original model. There are no conflicts of interest regarding the work.

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VII. Appendix

A. Optical Element Matrices

Using the $\mathcal{B}$ basis for the combined polarisation and propagation direction in section II, we can define each optical element:

BS

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i & 0 & 0 \\ i & 1 & 0 & 0 \\ 0 & 0 & 1 & i \\ 0 & 0 & i & 1 \end{bmatrix}$$

$$\begin{aligned} |H\rangle |X\rangle &\rightarrow \frac{1}{\sqrt{2}}(|H\rangle |X\rangle + i|H\rangle |Y\rangle) \\ |H\rangle |Y\rangle &\rightarrow \frac{1}{\sqrt{2}}(i|H\rangle |X\rangle + |H\rangle |Y\rangle) \\ |V\rangle |X\rangle &\rightarrow \frac{1}{\sqrt{2}}(|V\rangle |X\rangle + i|V\rangle |Y\rangle) \\ |V\rangle |Y\rangle &\rightarrow \frac{1}{\sqrt{2}}(i|V\rangle |X\rangle + |V\rangle |Y\rangle) \end{aligned}$$

Mirrors

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{aligned} |H\rangle |X\rangle &\rightarrow |H\rangle |Y\rangle \\ |H\rangle |Y\rangle &\rightarrow |H\rangle |X\rangle \\ |V\rangle |X\rangle &\rightarrow |V\rangle |Y\rangle \\ |V\rangle |Y\rangle &\rightarrow |V\rangle |X\rangle \end{aligned}$$

PBS1 & PBS2

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \end{bmatrix}$$

$$\begin{aligned} |H\rangle |X\rangle &\rightarrow |H\rangle |X\rangle \\ |H\rangle |Y\rangle &\rightarrow |H\rangle |Y\rangle \\ |V\rangle |X\rangle &\rightarrow i|V\rangle |Y\rangle \\ |V\rangle |Y\rangle &\rightarrow i|V\rangle |X\rangle \end{aligned}$$

QWP (First Pass)

QWP is placed at an angle θ from the horizontal polarisation axis on the first passage.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 \theta + i \sin^2 \theta & 0 & (1-i) \cos \theta \sin \theta \\ 0 & 0 & 1 & 0 \\ 0 & (1-i) \cos \theta \sin \theta & 0 & i \cos^2 \theta + \sin^2 \theta \end{bmatrix}$$

$$\begin{aligned} |H\rangle |X\rangle &\rightarrow |H\rangle |X\rangle \\ |H\rangle |Y\rangle &\rightarrow [(\cos^2 \theta + i \sin^2 \theta) |H\rangle + ((1-i) \cos \theta \sin \theta) |V\rangle] |Y\rangle \\ |V\rangle |X\rangle &\rightarrow |V\rangle |X\rangle \\ |V\rangle |Y\rangle &\rightarrow [((1-i) \cos \theta \sin \theta) |H\rangle + (i \cos^2 \theta + \sin^2 \theta) |V\rangle] |Y\rangle \end{aligned}$$

SBM

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

QWP (Second Pass)

QWP is placed at an angle $\pi - \theta$ from the horizontal polarisation axis on the first passage.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 \theta + i \sin^2 \theta & 0 & (i-1) \cos \theta \sin \theta \\ 0 & 0 & 1 & 0 \\ 0 & (i-1) \cos \theta \sin \theta & 0 & i \cos^2 \theta + \sin^2 \theta \end{bmatrix}$$

$$\begin{aligned} |H\rangle |X\rangle &\rightarrow |H\rangle |X\rangle \\ |H\rangle |Y\rangle &\rightarrow [(\cos^2 \theta + i \sin^2 \theta) |H\rangle + ((i-1) \cos \theta \sin \theta) |V\rangle] |Y\rangle \\ |V\rangle |X\rangle &\rightarrow |V\rangle |X\rangle \\ |V\rangle |Y\rangle &\rightarrow [((i-1) \cos \theta \sin \theta) |H\rangle + (i \cos^2 \theta + \sin^2 \theta) |V\rangle] |Y\rangle \end{aligned}$$

HWP

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & \sqrt{2} \end{bmatrix}$$

$$|H\rangle |X\rangle \rightarrow \frac{1}{\sqrt{2}}(|H\rangle |X\rangle + |V\rangle |X\rangle)$$

$$|H\rangle |Y\rangle \rightarrow |H\rangle |Y\rangle$$

$$|V\rangle |X\rangle \rightarrow \frac{1}{\sqrt{2}}(|H\rangle |X\rangle - |V\rangle |X\rangle)$$

$$|V\rangle |Y\rangle \rightarrow |V\rangle |Y\rangle$$

VWP

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\varepsilon} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$|H\rangle |X\rangle \rightarrow |H\rangle |X\rangle$$

$$|H\rangle |Y\rangle \rightarrow |H\rangle |Y\rangle$$

$$|V\rangle |X\rangle \rightarrow e^{i\varepsilon} |V\rangle |X\rangle$$

$$|V\rangle |Y\rangle \rightarrow |V\rangle |Y\rangle$$

A Classical Interpretation of Wheeler’s Delayed-Choice Double-Slit Experiment

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The underlying mysteries of quantum physics are closely tied to the model’s departures from classical light theory. While certain experiments may give the impression of being fundamentally quantum in nature, a deeper analysis often reveals that they can be understood through purely classical principles. We build upon our previous classical analysis of Wheeler’s ‘Delayed-Choice Double-Slit’ experiment by analysing the original work and considering potential causal flaws in the paper’s retrocausal argument.

I. Background

– Motivation –

THE very fundamental questions in the quantum theory of light aim to find an all-encompassing explanation for the behaviour of light. Although quantum mechanics is extremely successful in some of its justifications using rigorous mathematical frameworks, it leaves gaps in physical interpretations of certain phenomena such as wave-particle duality and lack of determinism in event outcomes. The source of many such gaps arrives from the ‘observer-dependence’ of measurements. More problematically, the concept of measurement is vague by essence and is, for example, generally defined as “*An irreversible act of amplification (revealing information about a system)*” [3] by the likes of Niels Bohr. Thus, one needs to thoroughly investigate measurements in order to understand whether quantum mechanics can indeed successfully explain a specific phenomenon. In this paper, we address a famous *gedanken* variation of the double-slit experiment raised by John Archibald Wheeler to understand *whether we can use purely classical interpretations to explain his “Double-Slit Delayed-Choice” experiment*.

– The Double-Slit Experiment –

The Youngian Double-Slit Experiment and its variations such as [figure 1](#) demonstrate wave-particle duality. Substantially, the experiment involves a coherent monochromatic light source that is adjustable in its wave packet emission (i.e. one can choose to shoot a short pulse of light in Gaussian form). The emitted light is incident on a barrier including two sufficiently narrow slits with width δ , as well as a screen positioned far away such that the distance between the barrier and screen $L \gg \delta$, recording the incident pattern of light. If the light source is fixed to shoot a short pulse of light, a double-slit interference pattern is observed ([figure 1a](#)). However, if one chooses to place a detector specifically after one of the slits, the incident pattern on the screen changes to two wide ‘Gaussian’ slit patterns ([figure 1b](#)). Classical and Quantum mechanics provide separate interpretations for this phenomenon. In the quantum framework, the emission is

treated as a *quanta of light* (photon) and travels through both slits simultaneously, with its superposition causing constructive/destructive interference to result in double-slit bright/dark spots. Once a measurement is made after one of the slits, the travelling path of the photon is determined and the state collapses to show the Gaussian pattern. Classically, the small packet of light is still treated as electromagnetic waves that diffract through both slits, revealing the expected double-slit pattern. Classical mechanics fails to provide a cohesive explanation to why there must be a difference in the detection pattern given whether there is a detector after one of the slits or not. The **which-path information (WPI)** is the information available regarding the path that light takes through either slit. WPI is correlated to the availability of the interference pattern and understanding the transformation of the WPI from none to all information is crucial to connecting the classical and quantum models which are sufficient but disjoint in their interpretations.

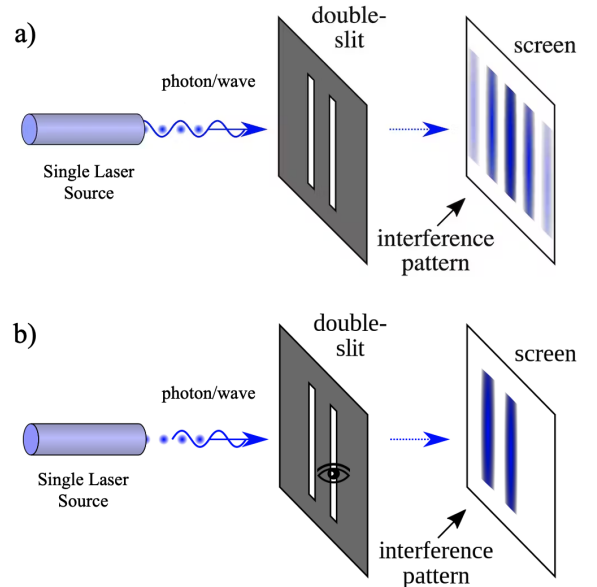


FIG. 1: Simplified schema of the DSE with **a)** no detector in place (WPI available) and **b)** detector in front of a slit (WPI unavailable) (*Image source: Johannes Kalliauer (2017)*)

– *Wheeler’s Experiments* –

In 1987, Wheeler published a series of thought experiments [3] that aimed to challenge our understanding of quantum mechanics by introducing ‘delayed choice’. In this paper, Wheeler proposed and analysed variations of Young’s double slit experiments with configurations that allowed for a time-dependent choice to affect the which-path information of light. Classically, the disturbance due to measurements of an ideal isolated system can be reduced in order to preserve the maximal information to some degree. In quantum mechanics, the probabilistic nature of results and especially the *Bell-Kochen-Specker Theorem* [2] prevents measurement disturbances from being arbitrarily minimised. Wheeler thus proposed a solution to the lack of system losslessness by introducing the ‘retrocausality’ of events. This naturally raised more challenging questions regarding the quantum nature of light.

– *The Thought Experiment* –

Figure 2 shows Wheeler’s original ‘Delayed-Choice Double-Slit Experiment’. In this setup, a narrow portal allows the emission of a pulse of monochromatic, coherent light. The beam is refracted at a lens that is covered by a screen that only allows a secondary emission from two slits. The propagated light hence moves towards the focal point where a plate is placed with a movable hinge (plate is positioned just after the focal point). The active/inactive (blocking/not blocking light) state of the plate corresponds to the delayed choice in the experiment. Much farther than the focal point two photodetectors, symmetric about the horizontal, are aligned to receive the beam of light. The detection modes are

1. **Plate active:** The position of pulse is measured across the plate instead of the photodetectors.
2. **Plate inactive:** The position of pulse is measured at the respective photodetectors.

Considering a single photon in the system, the active plate measures the probability distribution associated with a double-slit interference pattern. Given it is unclear whether the detected pulse has travelled through the upper or lower slit, Wheeler claims that this detection mode is associated with ‘no WPI’. However, if the plate is inactive and the photodetectors are engaged instead, then exactly one detector will be equally likely to register the pulse, thus revealing the ‘WPI’. Depending on the position of the two photodetectors, there will exist two detection peaks separated by a time delay due to the difference in path length from each slit. Wheeler interprets the first mode as “a single photon simultaneously travelling through both slits”, and the second mode as “photon travelling through only one slit”. Furthermore, Wheeler claims that the difference between the two modes

is also associated with distinct detection patterns and importantly, one may directly associate the existence of WPI with these detection patterns. The difference between the two detections is loosely inferred to as a duality in the state of light, where the first mode implies a *wave model* and the second mode implies a *particle model* of light.

According to Wheeler, a difficulty will arise when a delayed choice about the state of the plate is made *after* the light has traversed through the slits. As the choice is directly associated with a different interpretation of the model of light, Wheeler proposes that light behaves retro-causally, changing its behaviour before detection. He states:

“It is not a paradox that we choose what shall have happened after it has already happened... it is not a phenomenon until it is an observed phenomenon.” [4]

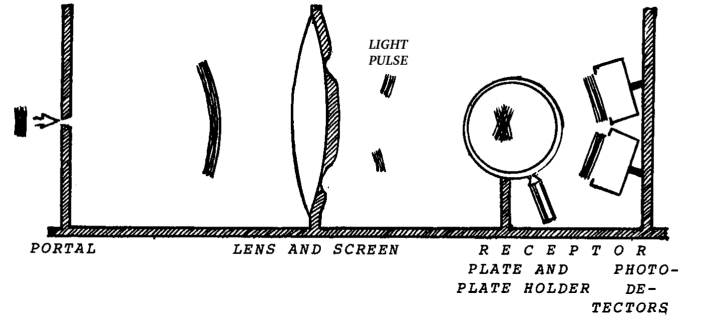


FIG. 2: Wheeler’s original Double-Slit Delayed-Choice setup [3]

– *Proposed Analogue* –

Upon investigating the basis of Wheeler’s thought experiment, we previously proposed an analogous model to the delayed choice [0]. We considered a ‘Split-Beam Delayed-Choice’ experiment summarised in figure 3 as a Mach-Zehnder interferometer variation. The proposed model, inspired by Pantaleoni’s [1] model on the delayed choice experiment, demonstrated that with certain configurations of interferometric apparatus, one can replicate Wheeler’s results with complete reliance on classical theory. We concluded that light’s choice is not classically well-defined but its nature is only manifested at the detectors once the WPI is already determined. It was also observed that the degree of WPI, that is the **maximal information** regarding the path of the light pulse, is adjustable and can vary. This result allowed for mitigation of retrocausality but

[0] [Link access to the paper](#)

raised a yet more challenging question: *Can one smoothly transform between a state of complete WPI, and no WPI?*

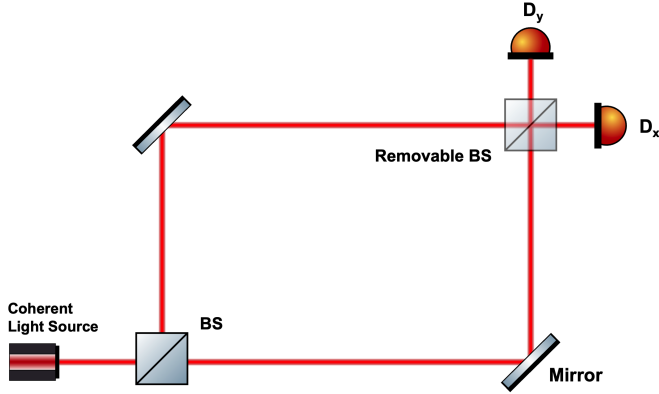


FIG. 3: A variation of Wheeler's thought experiment with the existence or lack of existence of the removable beam splitter determining the delayed-choice behaviour of light.

II. Approach & Results

– The Delayed-Choice Double-Slit –

We begin by constructing a model shown in [figure 4](#) similar to Wheeler's thought experiment in [figure 2](#), only relying on the classical wave theory of light. The coherent light source functions as described previously and the light through the slits is planar, propagating towards the focal point F . The twin photodetectors are aligned along a measurement arc which allows the apparatus to shift along this path while remaining symmetric about the horizontal. The intensity of light from the contribution of both light sources at each detector is derived using a position vector $\vec{r}$ where we wish to record the electric field. The light intensity then can be analysed under a Fraunhofer regime where the separation between the two slits $|\vec{s}_1 - \vec{s}_2| \ll \ell$, where ℓ is the horizontal distance between the slits and the centre of the measurement arc. Given the setup is symmetric, we can also consider that $|\vec{s}_1| = |\vec{s}_2|$. To preserve the 'pulse-like' nature of the emission, the light source emits a single Gaussian envelope with width τ , angular frequency ω , and $\mu_1(\vec{r}, t), \mu_2(\vec{r}, t)$ as the evolving widths of the Gaussian envelope perpendicular to the direction of propagation. $E_1(\vec{r}, t)$ and $E_2(\vec{r}, t)$ are the electric field amplitudes from the slits 1 and 2 respectively. $\vec{s}_1 + \vec{s}_2 = 0$ and define $\vec{p}_1 = \vec{d} - \vec{s}_1, \vec{p}_2 = \vec{d} - \vec{s}_2$ where $\vec{d}$ is the position vector of the focal point. Considering time-independent wave vectors $\vec{k}_1 = k\vec{p}_1$ and $\vec{k}_2 = k\vec{p}_2, k \in \mathbb{R}$, then the light amplitude can be described as

$$E_1(\vec{r}, t) = E_1 \exp\left(-\frac{(\vec{r} \cdot \hat{p}_1 - \omega t)^2}{2\tau^2}\right) \exp\left(\frac{(\vec{r} \cdot \hat{p}_1)^2 - |\vec{r}|^2}{2\mu_1(\vec{r}, t)^2}\right) \exp\left(i(\vec{k}_1 \cdot (\vec{r} - \vec{s}_1) - \omega t)\right) + c.c$$

$$E_2(\vec{r}, t) = E_2 \exp\left(-\frac{(\vec{r} \cdot \hat{p}_2 - \omega t)^2}{2\tau^2}\right) \exp\left(\frac{(\vec{r} \cdot \hat{p}_2)^2 - |\vec{r}|^2}{2\mu_2(\vec{r}, t)^2}\right) \exp\left(i(\vec{k}_2 \cdot (\vec{r} - \vec{s}_2) - \omega t)\right) + c.c$$

where $c.c.$ is the complex conjugate of the stated terms. Considering the intensities combined $I = (E_1 + E_2)^2$, then

$$I(\vec{r}, t) = E_0^2 \left[\exp\left(-\frac{(\vec{r} \cdot \hat{p}_1 - \omega t)^2}{2\tau^2}\right) \exp\left(\frac{(\vec{r} \cdot \hat{p}_1)^2 - |\vec{r}|^2}{2\mu_1(\vec{r}, t)^2}\right) \cos(\vec{k}_1 \cdot (\vec{r} - \vec{s}_1) - \omega t) + \exp\left(-\frac{(\vec{r} \cdot \hat{p}_2 - \omega t)^2}{2\tau^2}\right) \exp\left(\frac{(\vec{r} \cdot \hat{p}_2)^2 - |\vec{r}|^2}{2\mu_2(\vec{r}, t)^2}\right) \cos(\vec{k}_2 \cdot (\vec{r} - \vec{s}_2) - \omega t) \right]^2$$

– Predictions of Classical Analysis –

We analyse the thought experiment's results by changing two distinct variables. 1) The angle θ of deviation from the horizontal, and 2) the distance $|\vec{r}|$ from origin to the measurement arc. We consider distances $|\vec{r}| \approx |\vec{d}|$ (near focal point), $|\vec{d}| < |\vec{r}| < \ell$, and $\vec{r} = \ell$ where each measurement is made at angles $\theta = 0^\circ, 25^\circ, 50^\circ$. Hence we consider results for nine cases.

Near the focal point F ($|\vec{r}| \approx |\vec{d}|$) and when the detectors are overlapping ($\theta = 0$) the interference pattern and no WPI (complete overlap between light arriving from each slit) is observed. Once the detectors are not overlapping ($\theta = 25, 50$) a position dependant separation between the two path modes is visible where the difference between the modes is related to the amount of path difference of light from each slit, i.e. larger difference with larger angles as seen in [figure 5a](#). Importantly, there is a continuous transformation between the modes of maximum/minimum WPI with a certain overlap between the modes which can be arbitrarily decreases. The intensity curves for each photodetector is also similar and as the same result would be obtained for one photodetector if it was reflected about the horizontal axis. When the photodetectors are too close the distinction between light paths is very small and hence WPI is lost.

At distances $|\vec{d}| < |\vec{r}| < \ell$ further separation between the modes as well as decrease in overall intensity is observed ([figure 5b](#)). The ability to retrieve WPI depends more strictly on the detection time of the light pulse. Depending on the photodetector which registers the pulse, the time delay allows to understand which slit the pulse has traversed from, thus retrieving the WPI.

At the greatest distance $\vec{r} = \ell$, a near total separation of the two peaks, as well as the lowest intensity from all cases is observed. Noticing where $\vec{r} = \ell$ and $\theta = 25$ case (middle of [figure 5c](#)), one of the peaks approaches near no detection. This directly corresponds to Wheeler's delayed choice experiment where the setup is designed such

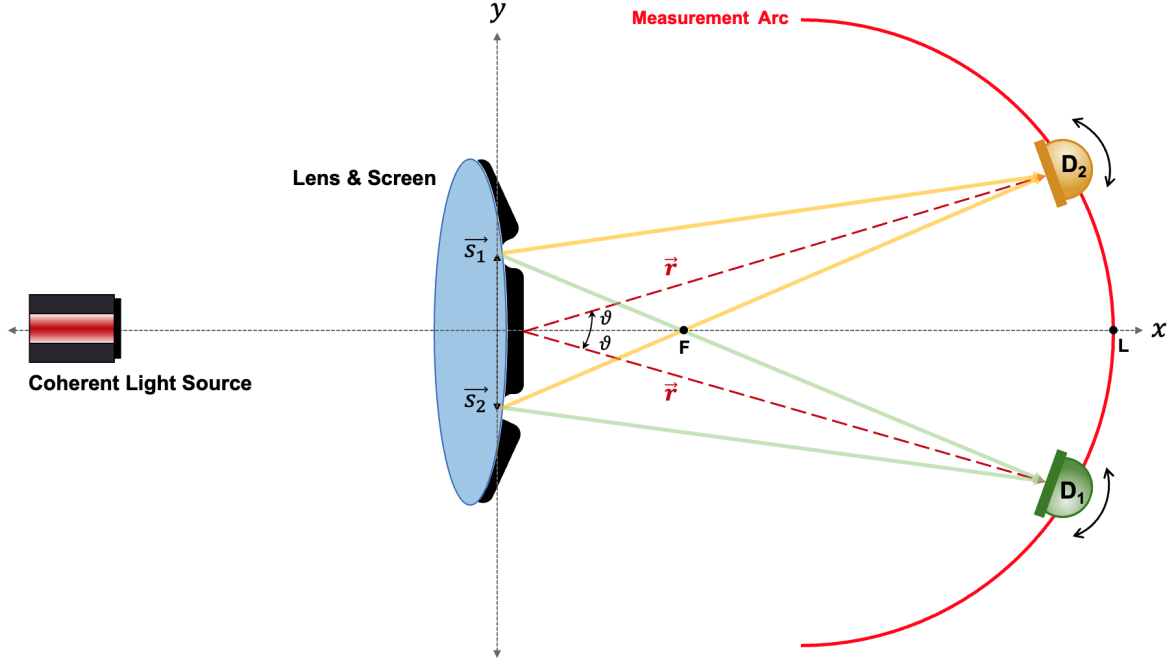


FIG. 4: Schema of our version of Wheeler's experiment. The positions of the detectors vary as θ changes. The origin is at the midpoint of the two slits.

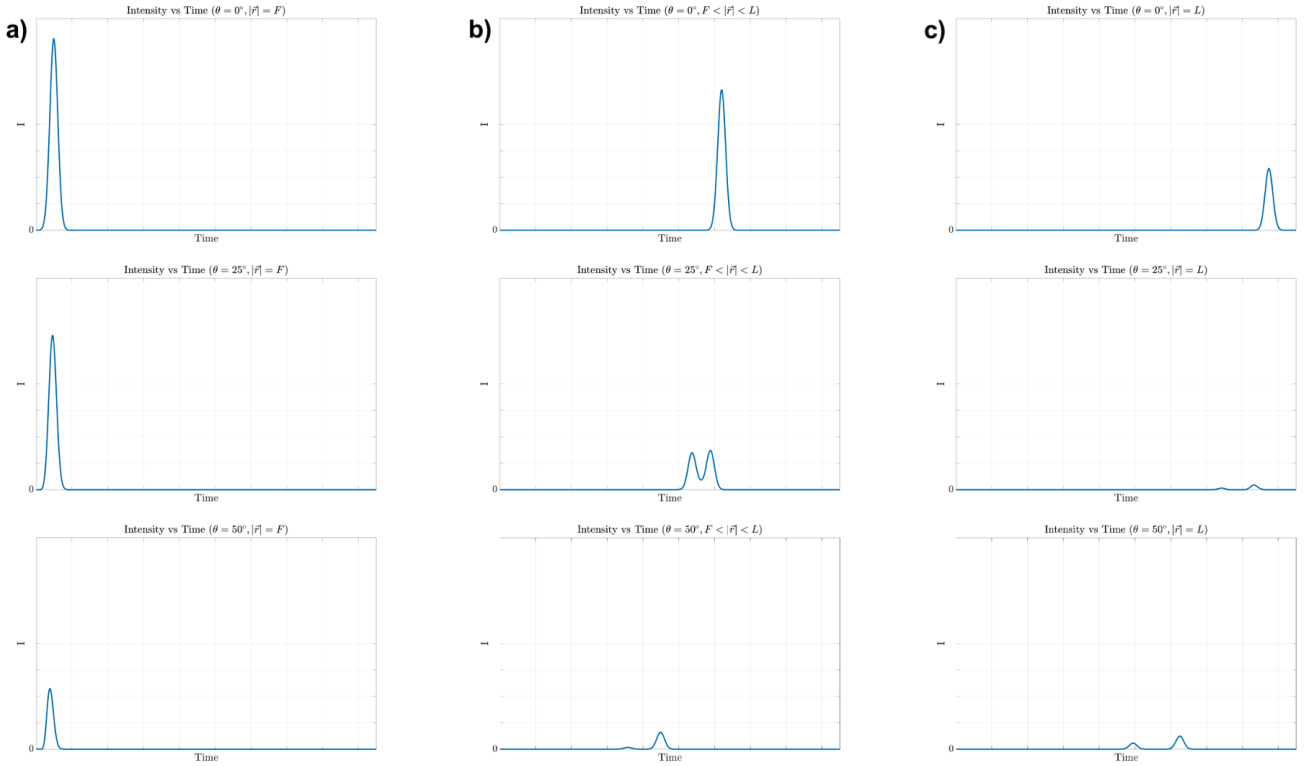


FIG. 5: The intensity-time plots of measurements from photodetectors at different measurement arcs and angles. **a)** measurement arc near focal point, **b)** measurement arc between focal point and ℓ , **c)** measurement arc at ℓ . The plots from each detectors are overlapping.

that the position of the photodetectors are fixed at this point. This is the maximal information state where WPI is known, whereas the case $\theta = 0$ corresponds to minimal information state which can be interpreted as measuring the interference when the plate is active.

– Double Slit Results –

In order to show a similarity between Wheeler's setup and the Youngian double-slit results, the results from our classical analysis must match the Youngian detection pat-

terns. As stated before, the double-slit interference pattern consists of bright/dark bands. Figure 6 shows our analysis where instead of shifting the photodetectors about a measurement arc, a vertical shift is made. Figure 6a to figure 6f show the detections in increasing vertical height. Clearly the detections oscillate in amplitude where figure 6a is the brightest spot, figure 6d is a completely dark spot, and figure 6f shows signs of moving back to a bright spot. The existence of this trend demonstrates the similarity between our results and the Youngian double-slit experiment.

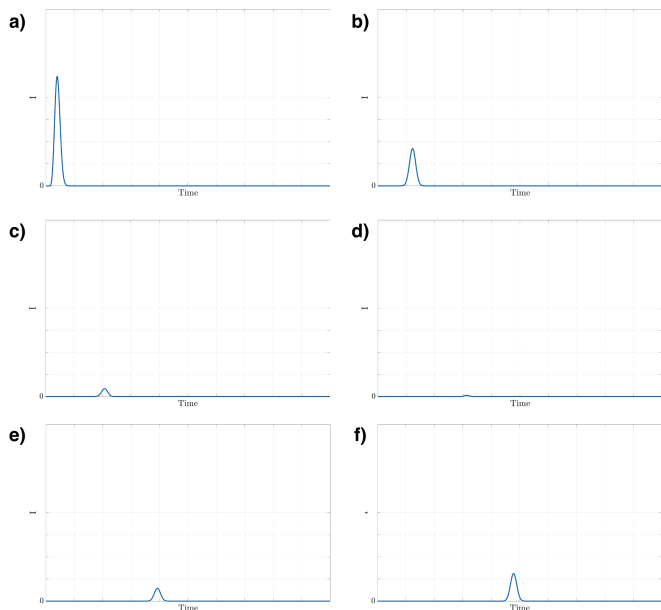


FIG. 6: Results for a double-slit interference pattern as detectors are shifted vertically. Graphs (a)-(e) correspond to positions further above from the focal point, starting from exactly on the focal point.

III. Discussion

– Insights –

Importantly, the relevant interference patterns are retrieved using our classical analysis to correspond to maximal/minimal information states. We find that gaining WPI denoted as determining which slit(s) light has traversed through cannot result in a state of maximal information. It is only at the extremal WPI and no WPI modes that the relevant association with the interference patterns may be observed. This means that one may choose to retrieve the a distribution of WPI results without disturbing the system. The implications of this are discussed in the next section.

Furthermore, it is crucial to understand that the current results demonstrate a continuous transformation between complete/no WPI modes. As discussed before,

each of these extremal WPI modes are interpreted as the particle/wave models of light respectively. Measurements with expanding region of minimal overlap correspond to complete WPI and hence a particle model, and the shrinking region of maximal overlap correspond to no WPI. Hence, a smooth shifting between WPI results implies a smooth change in the state of light which is not as expected. Provided that the setup remains to be lossless, one must strive to find an appropriate interpretation for this continuous transformation to ensure the overlap between the Gaussian peaks do not arise from classical uncertainties. Of course it is notable that this continuity is relatable to position-based classical interpretation of light where the emitted pulse is still treated as ‘*electromagnetic kicks*’.

– Wheeler’s Fallacy –

In a quantum mechanical interpretation of the experiment results, the act of retrieving WPI produces a system that is no longer in maximal information, i.e. measurement disturbs the system. As shown in figure 7, the double slit pattern is in correlation with no WPI mode if and only if light simultaneously traverses through the both slits, and the Gaussian pattern is in correlation with complete WPI mode if and only if light traverses through either slit (never at the same time). It is well known that the knowledge of ‘which-slit information’, that is whether light has traversed through both or either slits, is related to the detection patterns without the need to reduce the maximal information state of the system. Wheeler’s thought experiment claims that this which-slit information is also in association with the WPI modes as described before. Hence, changes in WPI modes affect the detection patterns and vice versa. Wheeler’s concern roots from the fact that a time-dependent choice in measuring this implication can result in a behaviour of light that can only be justified retrocausally. Using our classical interpretation, we suggest that this justification may be fallacious as the correlation between which-slit and which-path information requires some disturbance in the system by its nature (one cannot necessarily find maximal information about both of these at the same instance). The existence of measurement disturbance on one side of this three-way implication and lack of it on the other side as clearly seen in figure 7 implies that information flow can occur even though information can be lost which is fallacious. This is only possible through the use of a so called ‘*transparent detector*’ which conserves the WPI measurement disturbance and allows for a completely causal connection between which-slit and which-path information.

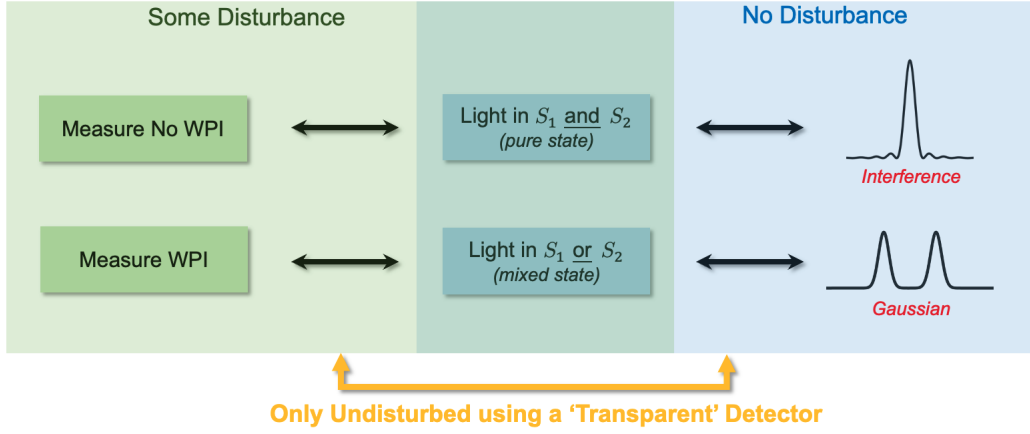


FIG. 7: Visualisation of the correspondence between WPI modes, Which-Slit information, and detection patterns. Measuring the left-hand-side disturbs the system while measuring the right-hand-side keeps the system lossless.

– Future Directions –

It would be of great interest to understand whether all variations of the delayed choice experiment as stated in Wheeler’s original paper are consistent with our analysis. In order to demonstrate a lack of particular quantum mystery in these series of experiments, each setup must be reduced to its abstract components and it must be ensured that these components are shared amongst all of the experiments. Furthermore, this reduction should be used in the future to simplify the basis of our past work on the proposed model to ensure the thought experiment itself is analogous to Wheeler’s work. It is only after this ‘isomorphism’ between the experiments that the ontology of the nature of light can be thoroughly discussed.

IV. Conclusion

In this paper, we classically analysed a famous case of Wheeler’s delayed-choice thought experiments and were able to reproduce the same correspondences between interference patterns and WPI modes. We constructed a closely matched model for the delayed-choice double-slit experiment and predicted the detection results for at least nine different cases. We determined the changes in WPI modes are continuous depending on the position of the photodetectors about the measurement arc which can be interpreted as a smooth change in wave/particle models of light. This is inconsistent with the quantum model of light and demonstrates a potential fallacy within Wheeler’s in-

terpretation of the original experiment. This is also reaffirmed through the discovery that the connection between WPI modes and detection patterns exist in states of maximal information and without information loss even after disturbing the system. Hence, we suggested that mysteries with the retrocausal structure of Wheeler’s interpretations can be replaced with a classical analysis of a similar system.

V. Acknowledgement

We would like to acknowledge Sahand Mahmoodian for providing their lecture notes for PHYS2921 (Optics). We also acknowledge Angela Karanjai’s kind efforts as the supervisor of the project for discussing and assisting with various ideas and most importantly for proposing the research idea.

VI. Contribution & Reflection

The authors worked separately and together in this project to address the aim. Computations, modelling, and other qualitative aspects were mostly done separately and the executive decision-making and major recollections were made during group meetings. The initial part of the project was spent on understanding various aspects of Wheeler’s experiments and the second half was spent on modelling the main experiment. There are no conflicts of interest regarding the work.

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Experimental Realisation of a Classical Analogy to the Quantum Eraser

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The foundations of quantum physics are closely tied to the model's departures from classical light theory. We consider a classical analogy to the Quantum Eraser experiment in the Mach-Zehnder interferometer setup. By manipulating polarized light, we emulate “which-path” information in a classical context. The study demonstrates that existence or erasure of interference patterns are in complete correspondence to light polarisation in each arm in the classical framework. Our results highlight the similarities and differences between classical wave behavior and quantum phenomena, offering insights into the fundamental principles of Quantum Physics.

I. BACKGROUND

Introduction

Quantum mechanics manifests paradigm-shifting revelations in areas where a classical theory of light cannot comprehensively describe the relevant phenomena. In the “classical” theory, light can be described as the propagation of linearly independent electric and magnetic wave components in space and time. The dependency of measurement outcomes on the observer's role in quantum mechanics shows fundamental discrepancies between the two theories. As such, many experiments have been designed to investigate quantum interpretations that appear to be different to that of the classical theory.

In the 1980s John Archibald Wheeler proposed a series of *Gedankenexperiments* to investigate the causal significance of the observer and measurement collapse [1]. A variation of his initial proposal to the ‘delayed’ choice double slit experiment inspired The “Quantum Eraser experiment” which was conceived by Scully and Drühl in 1982 [2]. They investigated how the act of measuring “**which-path**” information (the knowledge of the photon's trajectory) in a double-slit setup can affect the interference pattern observed with photons. The experiment demonstrated that the quantum interpretation is contingent on the knowledge of the observer, reaffirming the non-deterministic nature of quantum mechanics. We aim to gain an understanding of this experiment by implementing a classical analogy that utilises optical components in a Mach-Zehnder Interferometer setup. This allows us to simulate aspects of erasure in a classical setting and understand what the fundamental differences between the two theories are and whether any direct equivalences exist.

The Quantum Experiment

One famous variation of the Quantum Eraser experiment is implemented using Mach-Zehnder interferometry [3]. In this setup, emitted photons are probabilistically directed in different arms of the interferometer via beamsplitters. Upon measurement, when path information is known, the interference pattern vanishes, indicating particle-like behavior. If this information is “erased” or not known, then the interference pattern is restored, suggesting wave-like behavior [4]. It is important to note that presence or absence of an interference pattern depends on whether which-path information is measured and available rather than the retrocausality of light which is a common misconception of the experiment. The Quantum Eraser highlights the principle of complementarity in quantum mechanics, where particles exhibit either wave-like or particle-like properties depending on the experimental context or the measurements incurring in the setup.

From the quantum-mechanical point of view, photons have two states in the interferometer corresponding to their trajectory in each arm. The superimposed wavefunctions of these states can be measured to result in the photon striking the screen in locations where bright fringes could form if the experiment for the one photon system was repeated multiple times [5]. Measurement of one photon cannot reproduce the expected interference pattern, but the cumulative recording of multiple photons striking the screen can produce patterns identical to interference patterns similar to that of the Double Slit experiment [6] which is crucial to establishing a classical analogy to the Quantum Eraser experiment.

Classical Analogy

In classical context, many inherently quantum mechanical notions such as superposition and which-path information are not well-defined. Hence creating a classical analogy to the Quantum Eraser involves introducing optical principles such as light polarisation. Polarized light can be manipulated to make available or erase informa-

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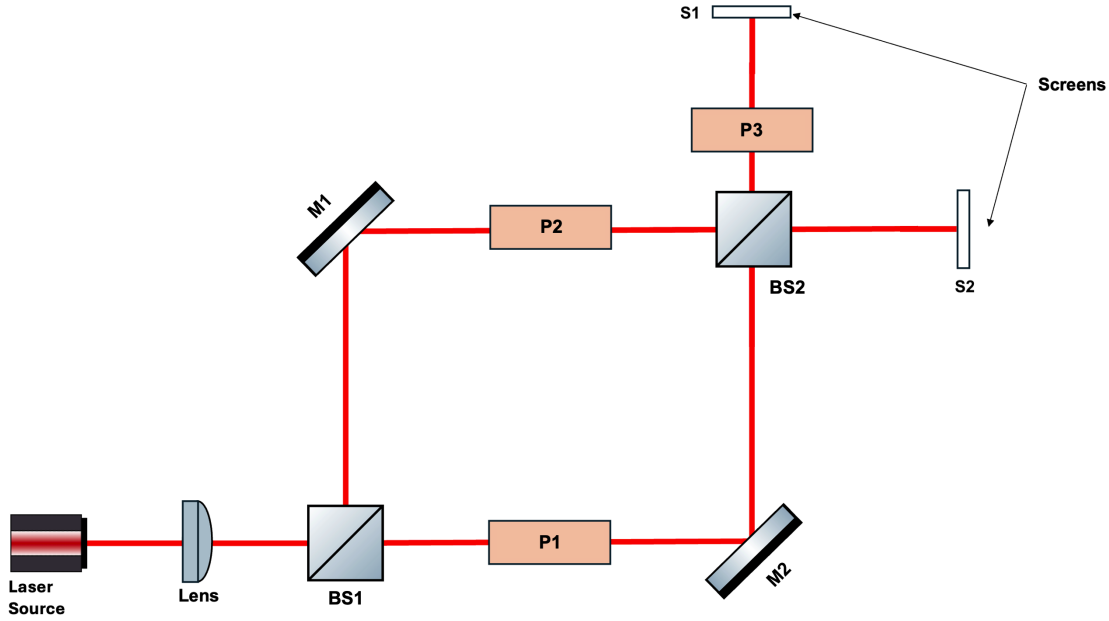


FIG. 1. Cartoon schema of the setup with each component labelled. The red beam represents the light beam in the setup. Note that $BS1$ and $BS2$ have opposing reflective sides.

tion as only light beams with similar polarisation can interfere. Additionally, measurement in the quantum context, which is a core part of this experiment is absent in essence. Although classical uncertainties exist within the measurement of the interference pattern, no measurement leading to wavefunction collapse is replicable. Also the experiment is substantially performed with single-photon light sources. It is then classically reasonable to predict that as a single photon passes through the interferometer, it can probabilistically traverse in an arm or the other and so due to the distinction of path information, interference pattern is lost. However, it is not classically reasonable to perform experiments with single photons. Instead we may consider a plane wave propagating in the z -direction of the form

$$\vec{E}(\mathbf{r}, t) = \begin{pmatrix} E_x \\ E_y e^{i\delta} \end{pmatrix} e^{i[\phi(z, t) + \phi_0]} \quad (1)$$

where

$$\phi(z, t) = (n\omega z/c) - \omega t. \quad (2)$$

Here n is the refractive index of the medium, c is the speed of light, δ is the relative phase difference between orthogonal amplitudes, and ϕ_0 is the global phase (which we can choose to ignore in our experiment). As clear from the continuous nature of $\vec{E}$, light is not longer quantised.

II. EXPERIMENTAL PROCEDURE

Mach-Zehnder Interferometer

The Mach-Zehnder setup of the experiment is shown in Figure 1. A coherent and monochromatic laser source emits a beam of light directed towards a 50:50 symmetric beamsplitter $BS1$. At the beamsplitter, light travelling from a less dense medium (air) and then reflecting upon a denser medium (glass) will gain a phase shift of 180° , while the transmitted light travelling from a more dense to a lesser dense medium gains no phase shift [7]. Upon this path separation, mirrors $M1$ and $M2$ guide these beams along their respective paths where two linear polarisers $P1$ (rotated to 0°) and $P2$ (rotated to 90°) change the polarisation in each arm. The beam then moves towards the second beam splitter $BS2$ which is crucially positioned with its reflective coating in opposition to that of $BS1$. This means that the light beam travelling from $M2$ can reflect and gain a phase shift, while the beam from the $M1$ path cannot gain any phase. The output beams are then projected on two flat screens $S1$ and $S2$ where the path to $S1$ includes another linear polariser $P3$ (rotated to 45°) [8]. Upon a complete setup, a plano-convex lens can be inserted before $BS1$ to magnify the final interference pattern shown on the screens.

The polarisation states determine whether the path information is retained or erased. If the polarisations are aligned, the which-path information is effectively erased, allowing the beams to interfere constructively or destructively. If the polarisations are relatively orthogonal

(e.g., 0° and 90°), the which-path information is preserved, preventing interference. Perfect alignment of the components results in interference patterns in the form of “Newton’s Rings” (See Appendix Fig 3). This particular pattern is due to the placement of the plano-convex lens from the interference of light waves reflecting from the top and bottom surfaces of a thin air film between a convex lens and a flat glass plate[9].

III. RESULTS & DISCUSSION

Complementarity

A two-dimensional intersection of the interference is depicted in Figure 2. It is thus visible that the combination of two complementary interference patterns results in a Gaussian envelope and even if path erasure occurs, this intensity will only distribute differently [3]. Without erasure (i.e. activating $P1$ and $P2$), the two arms are indistinguishable in polarisation. This leads to the complementary pattern of bright/dark interference rings seen in Appendix Figure 3. It is thus reasonable to state that the complementary interference in our setup is analogous to the cumulative recordings of single photons when no measurement is taken place in the quantum interpretation.

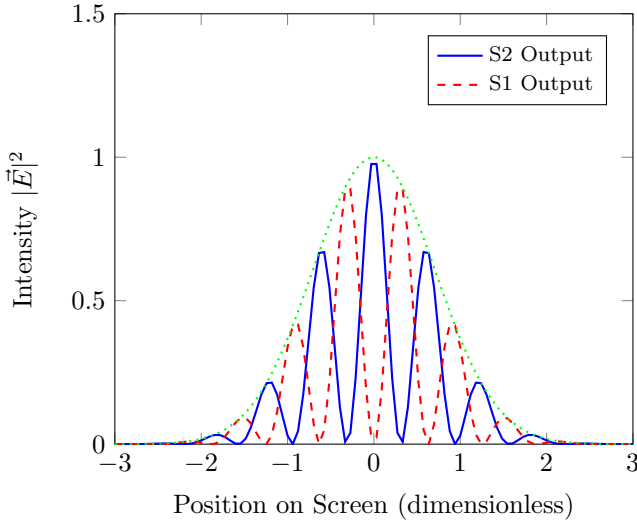


FIG. 2. Simulated plot of combined light intensity on $S1$ and $S2$.

The Erasure

Table I demonstrates the results for various settings where t (transmission) and r (reflection) denote the action of $BS1$ and $BS2$ (in order) on the light beams, showing the four possible paths of light. Hence tt and rr are detected on $S1$, while tr and rt are detected on $S2$. Interference patterns are visible on the screens only when

polarisations for each respective screen are the same. As stated before, this is clear for the case of no polarisers when complementarity is observed. Once $P1$ and $P2$ are active, the path information is erased and as seen in Table I (polarisation column 2), the polarisations of light beams that arrive at each screen are orthogonal and hence no interference occurs. The erasure is restored when $P3$ is active (polarisation column 3) where detections on $S1$ (paths tt and rr) match in polarisation and therefore interfere, while detections on $S2$ (paths rt and tr) are orthogonal and hence no interference occurs. It is thus reasonable to state that the existence of $P1$ and $P2$ polarisers is analogous to the erasure of path information, and activation of the $P3$ polariser is analogous to restoring the erased path information in the quantum interpretation.

In the quantum interpretation, the Heisenberg uncertainty principle implies that the more precise knowledge about the path information of the photon results in less precise knowledge about its momentum. If such information exists that reveals either the position or momentum of the particle, but then that information “erased” (so that the paths are indistinguishable), wave interference is again observed. Thus, the existence of the interference is purely dependent on measuring which path the photon has followed. In the classical framework, however, it is not a reasonable question to ask where the photon has travelled from, since light propagates in both arms at the same time. This argument also supports the lack of need for retrocausality since the act of measurement is not time-dependent in a classical system.

Path	Phase Relative Phase Shift	Polarisation		
		No Polarisers	No 45° Polariser	With All Polarisers
tt	0°	0°	0°	45°
rr	180°	0°	90°	45°
rt	180°	0°	90°	90°
tr	180°	0°	0°	0°

TABLE I. Phase shift and Polarisation angles for all 4 paths of the interferometer. Relative phase shift ignores the global phase accumulated due to reflection from mirrors.

Future Directions

Classical analogies lead to further understanding the fundamental concepts in Quantum Mechanics. To build upon this, future research could focus on investigating the Quantum Eraser experiment in long distance scales using satellites and probes. This can reinforce the importance of speed of light in such experiments where typically only small scales are considered. Additionally, it would be interesting to use light sources that resemble single-photon emitters such as light sources that emit Gaussian packets one-by-one to gain a closer analogy to the Quantum Eraser.

Finally, exploring potential applications in optical information processing and communication technologies could lead to innovations that leverage the principles demonstrated in this experiment.

IV. CONCLUSION

In this paper, we considered a classical analogy to the Quantum Eraser experiment and conducted an experiment in a Mach-Zehnder interferometer setup to explore the similarities and differences between the two interpretations. Using optical components, we replicated path-information and erasure results using the polarisation in different arms. The retrocausality of light was also refuted as the idea of time-dependent measurement is not well-defined in the classical framework. Overall it was found that although it is possible to replicate outputs that are equivalent in both interpretations, the fundamental physics that describe the processes are not fully aligned. Notably, the concept of

measurement which is uniquely defined in the quantum interpretation is not present in the classical analogy.

V. APPENDIX

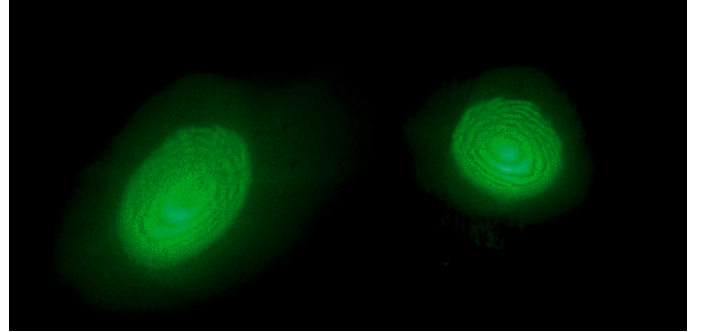


FIG. 3. the complementary interference captured from the experiment. The Newton Rings on the left corresponds to $S1$, and the other to $S2$.

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